13.3 The German Tank Problem

WORKSHEET

The question

STEP III 1998, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q14

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If M is the largest of n independent uniform values on $[0, N]$, what is $\mathrm{P}(M \leq m)$?

STAGE 1 Both are unbiased

Covers: the means of Z_1 and Z_2

Ask yourself The mean of Z_2 is quick: what is the mean of each X_i?

Pause and try For Z_1, find the distribution of the maximum M: when is $M \leq m$? Differentiate to get its density, then find $\mathrm{E}(M)$.

STAGE 2 Their variances

Covers: the variances

Ask yourself Find $\mathrm{E}(M^2)$ and so $\mathrm{Var}(M)$, then scale to get $\mathrm{Var}(Z_1)$. For Z_2, what is the variance of a single X_i? The X_i are independent.

STAGE 3 Which is better?

Covers: which to prefer

> **Ask yourself** Divide one variance by the other. How does the ratio behave as n grows? Can either estimator ever be smaller than a serial number you have actually seen?

What's really going on

Sufficiency

Why is the maximum so much better? Given the largest value M, the other values are just uniform on $[0, M]$, whatever N is: they carry no further information about N. The maximum is a *sufficient statistic*. A theorem of C. R. Rao and David Blackwell (1945–47) says that any unbiased estimator can be improved, or at least not made worse, by replacing it with its expected value given a sufficient statistic, and here that turns Z_2 into Z_1. The *Lehmann–Scheffé theorem* then shows that Z_1 is the unbiased estimator with the smallest possible variance. Notice too that the variance of Z_1 falls like $\frac{1}{n^2}$, faster than the $\frac{1}{n}$ of most estimators: that happens because N is an endpoint of the distribution, and the data pile up against it.

The discrete version

Real serial numbers are whole numbers $1, 2, \dots, N$, sampled without replacement. The best unbiased estimate is then $\hat{N} = m + \frac{m}{k} - 1$, where m is the largest of k observed serial numbers: the maximum plus the average gap between observations. Using estimates of this kind, based mainly on the serial numbers of gearboxes and wheels, the Allied statisticians estimated German production at about 246 tanks a month, against the 245 later found in German records and the 1400 suggested by intelligence reports.

Serial numbers in history

The method has been used many times since, for example by analysts estimating how many units of a new consumer product had been made from the serial numbers of the first few sold. Manufacturers learned the lesson: many now encode serial numbers so

that they cannot be read as a simple sequence. Richard Ruggles and Henry Brodie described the wartime work in the *Journal of the American Statistical Association* in 1947.

Try this next

1. **Mean squared error.** The biased estimator M has $\mathrm{E}(M-N)^2 = \mathrm{Var}(M) + \big(\mathrm{E}(M) - N\big)^2$. Compute it and compare with $\mathrm{Var}(Z_1)$. Find the constant c that minimises $\mathrm{E}(cM - N)^2$.

2. **The minimum too.** Show that $(n+1)\min\{X_i\}$ is also an unbiased estimator of N, and find its variance. Is it any good?

3. **The discrete formula.** For serial numbers $1, \dots, N$, sample k without replacement, with maximum m. Show that $\mathrm{P}(\max \le m) = \binom{m}{k} / \binom{N}{k}$, and verify that $m + \frac{m}{k} - 1$ is unbiased when $k = 1$.

4. **Tanks.** If 5 captured tanks have serial numbers 19, 40, 42, 60, 98, estimate the number of tanks made.

Further reading

- **Richard Ruggles & Henry Brodie, "An empirical approach to economic intelligence in World War II"**, *Journal of the American Statistical Association* 42 (1947). The original account.
- **John Rice, *Mathematical Statistics and Data Analysis*** (Duxbury), on bias, variance and sufficiency.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican).

The full guided solution is in Section 13.3:
https://beyondstep.keithmcnulty.org/ch13/german-tanks.html